BRAUDE
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INTELLIGENT CV SCREENING AND CANDIDATE RANKING SYSTEM

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The Problem :

The rapid growth in job applications makes manual resume screening time-consuming, inconsistent, and prone to bias.

Project Goal :

The goal of this project is to provide an automated and structured approach to candidate screening that supports recruiters in evaluating and ranking candidates consistently based on job-specific criteria.

Technologies used:

- **Frontend**: React (JavaScript) - used to build a component-based, single-page application that presents candidate and job data dynamically to recruiters.
- **Backend**: Node.js + Express - used to implement the server-side application, manage RESTful APIs, handle file uploads, and control the core business logic of the system.
- **Database**: MongoDB - used to store candidates, job postings, and extracted resume data in a flexible document-based structure.
- **Authentication**: JWT - used to authenticate recruiters and protect API routes by validating user identity and access permissions.
- **AI Tool**: OpenAI GPT-4o-mini - used to analyze uploaded resumes, extract structured information, and support automated candidate evaluation.
- **Ranking Algorithm**: K-Nearest Neighbors (KNN) - used to calculate candidate similarity and generate ranking scores based on resume features and job requirements.

Functional Requirements	Non-Functional Requirements
The system allows recruiters to register and log in securely.	The system provides a clear, intuitive, and user-friendly interface.
The system enables the creation and management of job postings that include evaluation criteria.	The system ensures data security and protection of sensitive candidate information.
The system allows uploading candidate resume files.	The system provides consistency in the evaluation process for all candidates.
The system performs automatic analysis of resumes and extracts relevant information.	The system supports handling resumes in different formats and unstructured layouts.
The system calculates a score and ranks candidates according to job requirements.	The system allows future extensibility of ranking algorithms and candidate evaluation methods.
The system displays a ranked list of candidates along with a score breakdown for each candidate.	The system ensures reliable performance and reasonable response times when processing resumes and generating candidate rankings.
The system allows viewing a detailed candidate profile.	The system is designed with modular architecture to improve maintainability and simplify future updates and enhancements.

Project Metrics and Goal Achievement

- **Candidate Ranking Generation –** The system generates a ranked list of candidates for each job position based on resume analysis and predefined evaluation criteria. Each candidate receives a score that enables comparison between applicants.
- **Consistency of Evaluation –** All candidates are evaluated using the same weighted criteria, ensuring uniform and repeatable assessment regardless of resume format or wording. This reduces subjective decision-making.
- **Resume Processing Capability –**The system processes unstructured resume files by converting them into plain text and extracting relevant information, enabling effective handling of real-world resume data.
- **Decision Support for Recruiters –** The platform presents ranking results along with score breakdowns, supporting informed and efficient hiring decisions.

Future Enhancements

- Interview scheduling directly from the system with candidate self-selection of date and time
- Saving candidates for future opportunities and long-term follow-up
- Internal notes and activity history for improved candidate evaluation and tracking

System Overview :

User Authentication → Users authenticate using encrypted passwords and token-based access. Authorization ensures users can access only their own jobs and candidates.

Dashboard Overview → After login, users access a dashboard that presents an aggregated overview of jobs and candidates using summary statistics and charts.

Job Creation → Recruiters define job details and weighted evaluation criteria that must total 100%.

Resume Upload and Processing → Candidate resumes are uploaded in several formats, Text is extracted from each file and stored.

AI-Powered Data Extraction → Extracted resume text is analyzed using OpenAI GPT-4o-mini, producing structured candidate data and per-criterion scores with explanations.

Candidate Ranking Algorithm → Candidates are ranked using a weighted KNN approach by comparing each candidate to an ideal profile. The distance is calculated as:

$$distance = \sqrt{\left(\sum\left(weight * (normalizedValue - 1.0)^2\right)\right)}$$

The final score is computed as:

$$overallscore = (1 - distance)*100$$

Candidates are ranked based on their overall score.

Smart Re-ranking and Quick Re-rank → When job requirements change, affected candidates are re-evaluated, while weight-only changes trigger a fast re-ranking using existing data.

Candidate Management → The system displays detailed candidate information and allows status updates, bulk actions, and secure deletion.

Challenges And Solutions:

- **Unstructured resumes** - All files are converted to plain text for consistent analysis.
- **Language and terminology variations** -Text is analyzed flexibly to handle errors and different expressions of the same skills.
- **Subjective evaluation** - Weighted evaluation criteria ensure consistent and objective scoring.
- **Lack of training data** - Candidates are compared to an ideal profile without relying on labeled datasets.
- **Result transparency** - Rankings include clear, criterion-level explanations.
- **Data security** - Access to sensitive candidate data is restricted to authorized users.

System Architecture :